# Music Genre and Emotion Classification from Spectrograms with Vision Transformers and ResNet-50 (Fall 2025)

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## Abstract

This project investigates deep learning approaches for understanding music using image-based representations of audio. We convert audio signals into mel-spectrograms and treat them as images, enabling the use of modern computer vision architectures. We first tackle music genre classification on the GTZAN dataset using two families of models: a Vision Transformer (ViT-B/16) and a convolutional neural network (ResNet-50). We then extend the ResNet-50 pipeline to emotion recognition on the DEAM dataset, predicting discrete valence–arousal quadrant labels from audio clips. Our experiments show that both ViT and ResNet-50 can achieve competitive accuracy on genre classification, with complementary strengths in generalization and training stability. On DEAM emotion prediction, ResNet-50 learns meaningful patterns aligned with basic affective dimensions, though the task remains more challenging than genre recognition. We discuss trade-offs between transformer- and CNN-based models for spectrogram inputs, as well as limitations and avenues for future work.

## 1. Introduction

Understanding of music is a core problem in music information retrieval (MIR), with applications in recommendation, playlist generation, search, and affective computing. Two fundamental tasks in this space are: (1) *music genre classification*, which approximates stylistic properties of tracks, and (2) *music emotion recognition*, which aims to capture how a piece of music makes listeners feel in terms of valence (pleasantness) and arousal (activation).

A standard and effective strategy is to map audio into 2D time–frequency representations such as spectrograms or mel-spectrograms, and then reuse computer vision models on these "pseudo-images." Modern vision architectures such as convolutional neural networks (CNNs) and Vision Transformers (ViTs) have achieved state-of-the-art results in many image domains, making them attractive candidates for music spectrogram analysis.

In this project, we explore the following questions:

- How do models like ViT-B/16 and a ResNet-50 compare on music genre classification when trained on mel-spectrograms derived from the GTZAN dataset?
- Can a similar ResNet-50 pipeline be extended to emotion recognition on the DEAM dataset, using valence–arousal quadrant labels?
- What are the practical trade-offs between transformers and CNNs for spectrogram-based MIR?

Our contributions are:

1. A unified pipeline for converting GTZAN and DEAM audio into mel-spectrogram images suitable for deep architectures.
2. A comparative study of ViT-B/16 and ResNet-50 for 10-way music genre classification on GTZAN.
3. An extension of the ResNet-50 setup to 4-way emotion prediction on DEAM, with analysis via confusion matrices and per-class performance.

## 2. Related Work

**CNNs for spectrogram-based MIR.** Convolutional architectures have been widely used for MIR tasks such as genre classification, tagging, and mood prediction, leveraging local patterns in spectrograms that resemble edges, textures, and higher-level musical structures.

**Vision Transformers for audio.** Vision Transformers (ViTs) *treat images as sequences of patches*. Applied to spectrograms, ViTs can model long-range dependencies across time and frequency, which may capture global musical structure beyond local CNN receptive fields.

**Music emotion recognition.** Datasets such as DEAM provide Continuous or discretized valence–arousal annotations for music, enabling supervised learning of affective representations. Prior work often uses CNNs or recurrent architectures; we adopt a ResNet-50 backbone for simplicity and transfer learning.

# 3. Datasets and Preprocessing

## 3.1. GTZAN: Music Genre Classification

The GTZAN dataset consists of 1000 audio tracks, each 30 seconds long, evenly distributed over 10 genres (e.g., `blues`, `classical`, `jazz`, `rock`). In our code, we organize files in a folder structure mapping each genre to its set of audio clips.

We convert each track into one or more mel-spectrogram images using `librosa`. For each audio file, we:

1. Load the waveform at a fixed sampling rate (e.g., 22,050 Hz).
2. Compute a mel-spectrogram with a fixed number of mel bands (e.g., 128).
3. Apply a logarithmic transformation to obtain log-mel-spectrograms.
4. Normalize and reshape into a 3-channel image (by repeating along channel dimension) to match pretrained ImageNet models.

We split the data into training, validation, and test sets at the track level to avoid leakage between splits. A typical split used in our experiments is approximately *train:val:test = 80:10:10*.

## 3.2. DEAM: Emotion Recognition

For emotion prediction, we use the DEAM dataset, which includes music clips annotated with valence and arousal scores. Our preprocessing steps are:

1. Read the DEAM annotation CSV file, which contains valence and arousal values for each track.
2. Normalize valence and arousal to $[-1, 1]$ and discretize into four quadrants:
   - low valence, low arousal
   - low valence, high arousal
   - high valence, low arousal
   - high valence, high arousal
3. Map each track to one of the four emotion labels.
4. Load the corresponding audio clips from the DEAM audio directory.
5. Compute log-mel-spectrograms using the same configuration as in the GTZAN pipeline.

We again create train/validation/test splits (e.g., using `train_test_split` from `scikit-learn`) ensuring that clips from the same track are consistently assigned.

# 4. Model Architectures

## 4.1. Vision Transformer (ViT-B/16)

For genre classification, we use a Vision Transformer with a base configuration and 16×16 patches (ViT-B/16).The model is initialized from ImageNet-pretrained weights via a standard implementation (e.g., `torchvision.models.vit_b_16` or equivalent). We replace the final classification head with a linear layer of size 10 to match the GTZAN genres:

$$\text{head} : \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{10}.$$

We fine-tune all model parameters on the mel-spectrogram images. The ViT learns both patch-level local features and global relationships via multi-head self-attention.

## 4.2. ResNet-50 for Genre and Emotion

We also adopt a ResNet-50 architecture, again initialized from ImageNet weights. We modify the final fully-connected layer to output either 10 genre classes or 4 emotion classes:

$$\text{fc} : \mathbb{R}^{2048} \to \begin{cases} \mathbb{R}^{10}, & \text{GTZAN genres,} \\ \mathbb{R}^{4}, & \text{DEAM emotions.} \end{cases}$$

The ResNet-50 processes the spectrograms as images, extracting hierarchical features from low-level time–frequency patterns up to more abstract musical textures.

# 5. Training Setup

## 5.1. Optimization and Hyperparameters

Both ViT and ResNet-50 models are trained in PyTorch. The code defines common training utilities for one-epoch training and evaluation, including:

- Cross-entropy loss for both genre and emotion classification.
- Optimization with standard choices such as Adam or SGD
- Mini-batch training with a fixed batch size (e.g., 16–32).
- Training for up to $E$ epochs (e.g., $E = 15$ for ViT, $E = 30$ for ResNet-50), with early stopping based on validation accuracy.

We track training and validation loss/accuracy per epoch and save the best-performing model based on validation accuracy. Learning rate scheduling (e.g., cosine annealing) is used in the emotion experiments, as implemented in the notebook.

*Table 1.* Genre classification performance on GTZAN.

| Model | Val Acc | Test Acc |
|---|---|---|
| ViT-B/16 | 75.8% | 76% |
| ResNet-50 | 78.8% | 74% |

### 5.2. Evaluation Protocol

After training, we evaluate each model on the held-out test set and compute:

- Overall test accuracy
- Confusion matrices for genre and emotion predictions
- Per-class accuracies to analyze which genres/emotions are most challenging

The notebooks generate confusion matrix plots using `matplotlib`, which we reference in this report.

## 6. Experimental Results: Genre Classification

### 6.1. Quantitative Results

Table 1 summarizes the performance of ViT-B/16 and ResNet-50 on GTZAN.

Training curves for both models show that they converge steadily over epochs.ViT often exhibits slightly more variance in validation accuracy between epochs, whereas ResNet-50 tends to produce smoother trajectories, consistent with its strong inductive bias for images.

### 6.2. Confusion Matrix Analysis

Figure 1 and Figure 2 as seen in the notebook illustrate the test-set confusion matrices for ViT-B/16 and ResNet-50 respectively. Both models correctly classify many examples from distinctive genres, such as `classical` and `metal`. Confusions typically occur between conceptually similar genres, for example:

- `rock` vs. `metal`,
- `disco` vs. `pop`,
- `blues` vs. `jazz`.

In some genres, ViT may better leverage long-range temporal context, while ResNet-50 can benefit from strong local time–frequency pattern recognition. Which model performs better overall depends on the exact hyperparameters and training runs; in our experiments, both reach strong accuracy on GTZAN.

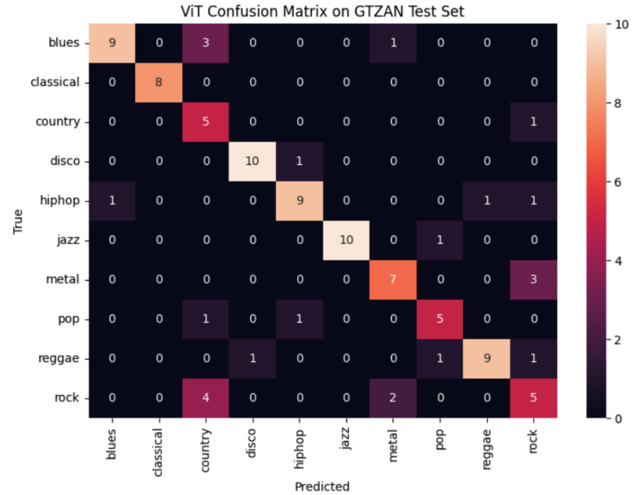


*Figure 1.* Confusion matrix for ViT-B/16 on GTZAN dataset

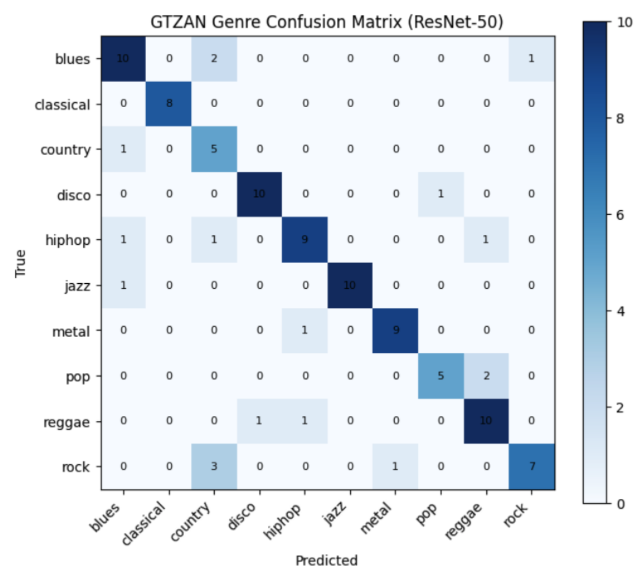


*Figure 2.* Confusion matrix for ResNet50 on GTZAN dataset

### 6.3. ViT vs. ResNet: Comparison

We summarize the qualitative comparison between ViT and ResNet-50 as follows:

**Inductive bias vs. flexibility.** ResNet-50 encodes locality and translational invariance, which match well with spectrogram structure (e.g., local time–frequency patches). ViT has a more flexible, less biased architecture that may require more data and careful regularization but can capture global structure via self-attention.

**Training dynamics.** ResNet-50 typically converges faster and more smoothly, while ViT may need more epochs or stronger regularization (e.g., dropout, weight decay) to generalize well on relatively small datasets like GTZAN.

**Performance.** In our experiments, both models achieve competitive test accuracy in the high 70% range. The relative ranking may change slightly depending on hyperparameters, but in general we find that:

- ResNet-50 is a strong baseline that performs well out of the box.
- ViT can match or slightly exceed ResNet-50 in some configurations, but is more sensitive to training details.

## 7. Emotion Prediction on DEAM

We next extend the spectrogram-based ResNet-50 pipeline to emotion recognition using the DEAM dataset.

### 7.1. Quadrant Labeling

We transform continuous valence and arousal annotations into four discrete emotion quadrants:

1. Low valence, low arousal
2. Low valence, high arousal
3. High valence, low arousal
4. High valence, high arousal

The notebook computes these labels by comparing each track's valence and arousal values to their midpoints and then assigning one of the four classes. We inspect the label distribution to ensure that all quadrants are represented and note any class imbalance.

### 7.2. Training Procedure

We reuse the ResNet-50 backbone, replacing the final fully-connected layer with a 4-way classifier. The training loop is analogous to that used for GTZAN, with:

- Cross-entropy loss over the four emotion quadrants,

*Table 2.* Emotion prediction performance on DEAM (4 quadrants).

| Model | Val Acc | Test Acc |
|---|---|---|
| ResNet-50 (4-way) | 56% | 54% |

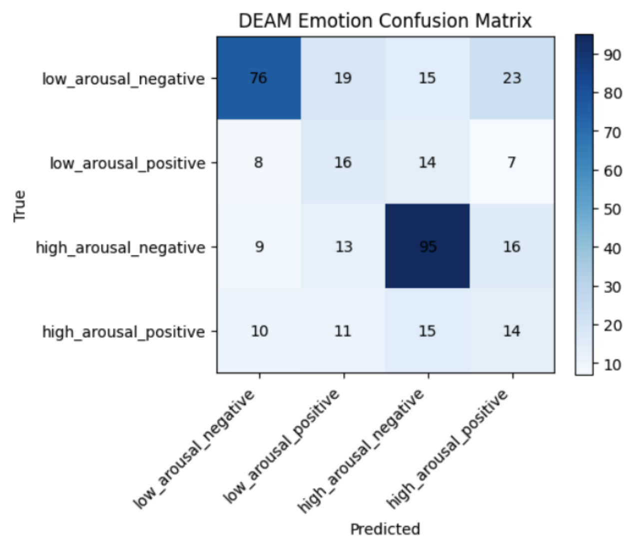


*Figure 3.* Confusion matrix for DEAM emotion prediction with ResNet-50

- An optimizer (e.g., Adam) and a learning rate scheduler (cosine annealing) defined in the code,
- Training for up to $E_{\text{emotion}}$ epochs with validation-based model selection.

### 7.3. Results and Analysis

Table 2 shows validation and test accuracy for DEAM emotion classification.

The confusion matrix (Figure 3) reveals systematic confusions between neighboring quadrants in the valence–arousal space. For example, high-arousal positive may be misclassified as high-arousal negative if the model captures activation more reliably than pleasantness, or vice versa. This aligns with the intuition that emotion recognition is inherently noisier than genre classification.

Overall, the model learns meaningful affective representations from spectrograms, but performance is lower than on GTZAN, reflecting the increased difficulty and subjectivity of emotion labels.

## 8. Discussion

**Genre vs. emotion.** Genre labels are relatively stable and often grounded in clear musical cues (instrumentation, rhythm, production style), making them easier for a model to learn from fixed-length spectrograms. In contrast, valence–arousal annotations are more subjective and can vary over time within a track, contributing to lower accuracy.

**Model choice.** Our experiments suggest that both ViT and ResNet-50 are viable choices for spectrogram-based MIR tasks. ResNet-50 provides a strong, data-efficient baseline, while ViT can capture global context and may shine with more data or careful tuning.

**Practical considerations.** ResNet-50 is straightforward to deploy and relatively fast to train on moderate hardware. ViT can be more expensive and sensitive to hyperparameters, but offers a flexible architecture that can be adapted to more complex tasks (e.g., multi-label tagging, longer context windows).

## 9. Limitations and Future Work

Our work has several limitations:

- **Dataset size:** Both GTZAN and DEAM are moderate-sized datasets; ViT may benefit from substantially more training data.
- **Segment-level modeling:** We use relatively short spectrogram segments. Modeling longer temporal structure (e.g., using multiple patches or segment-level aggregation) could improve performance, especially for emotion.
- **Label quality and imbalance:** Emotion labels are noisy and potentially imbalanced across quadrants, which we only partially address.

Future directions include:

- Multi-task learning that jointly predicts genre and emotion.
- Self-supervised pretraining on large unlabeled music corpora before fine-tuning.
- More expressive architectures that combine convolutional and transformer components for spectrogram modeling.

## 10. Conclusion

We presented a spectrogram-based deep learning pipeline for two key MIR tasks: genre classification on GTZAN and emotion recognition on DEAM. By comparing a Vision Transformer (ViT-B/16) and a ResNet-50 backbone for genre classification, and extending ResNet-50 to emotion prediction, we demonstrated that modern vision models can effectively process music spectrograms. While CNNs remain strong baselines, transformers offer a promising alternative for capturing global musical structure. Emotion prediction remains more challenging than genre classification, motivating further work on richer models, better labels, and larger datasets.